

1. (a) For all $\omega \in \Omega$, $X_n(x) = 1$ for large n (when $\frac{1}{n} < \omega$). This proves $X_n \rightarrow X$ a.s. (P).
 (b) For any $\epsilon > 0$, $n \geq 2$, $P(|X_n - X| > \epsilon) = \frac{1}{n} \rightarrow 0$ as $n \rightarrow \infty$.
 (c) $E(|X_n - X|^p) = (n-1)^p \cdot \frac{1}{n}$. So, $X_n \rightarrow X$ in L^p for $p \in (0, 1)$ but does not converge for $p \geq 1$.
 (d) $\{X_n\}$ is not UI, since for each $t > 1$, $E(|X_n|I_{\{|X_n|>t\}}) = n \cdot \frac{1}{n} = 1$ for n large ($n > t$). So for each $t > 0$, $\sup_{n \geq 1} E(|X_n|I_{\{|X_n|>t\}}) = 1$. Hence, taking the limit $t \rightarrow \infty$, the limit is not 0.
2. (a) With $g_n = f - f_n \geq 0$, apply Fatou's Lemma to get $\liminf_{n \rightarrow \infty} \int g_n d\mu \geq \int (\liminf_{n \rightarrow \infty} g_n) d\mu$. Hence, $\int f d\mu - \limsup_{n \rightarrow \infty} \int f_n d\mu \geq \int f d\mu - \int (\limsup_{n \rightarrow \infty} f_n) d\mu$. Now using the fact that $f \in L^1$, we get the result.
 (b) Using Cauchy Schwarz, we get $1 = \int f^{\frac{1}{2}} \left(\frac{1}{f^{\frac{1}{2}}} \right) d\mu \leq \left(\int f d\mu \right)^{\frac{1}{2}} \left(\int \frac{1}{f} d\mu \right)^{\frac{1}{2}}$. Squaring both sides yields the result.
 (c) Choose $A = \{f > g\} \in \mathcal{F}$. Then $0 = \int_A (f - g) d\mu = \int (f - g) \mathbf{I}_{\{(f-g)>0\}} d\mu$ implies $\mathbf{I}_{\{(f-g)>0\}} = 0$ a.e. (μ) or $\mu(f > g) = 0$, i.e. $f \leq g$ a.e. (μ). Similarly, one can argue $f \geq g$ a.e. (μ), which proves $f = g$ a.e. (μ).
3. (a) $\mu \perp \nu \Rightarrow \exists B \in \mathcal{F}$ s.t $\nu(B) = \mu(B^c) = 0$. Since $\mu \ll \nu$ and $\nu(B) = 0$, then $\mu(B) = 0$. So $\mu(B) = \mu(B^c) = 0 \Rightarrow \mu(\Omega) = 0$. Thus $\forall A \in \mathcal{F}, 0 \leq \mu(A) \leq \mu(\Omega) = 0 \Rightarrow \mu(A) \equiv 0, \forall A \in \mathcal{F}$, i.e. $\mu \equiv 0$.
 (b) Let $B = \mathbb{N} \cup \{0\} = \mathbb{N}_0$, then $3\mu_2(B^c) = 0 = \nu(B)$, i.e. $3\mu_2 \perp \nu$. Choose $\mu_{3s} = 3\mu_2, \mu_{3a} = \mu_1$ and $\mu_3 = \mu_{3a} + \mu_{3s}$. Now, since $\mu_{3a}(A) = \int_A \phi_{(-1,1)} d\nu$ we have $\frac{d\mu_{3a}}{d\nu} = \phi_{(-1,1)} =$ density function of $Normal(-1, 1)$.
 (c) Since $\frac{d\mu_{3a}}{d\nu} = \phi_{(-1,1)}$, we have, $\int g d\mu_{3a} = \int g \phi_{(-1,1)} d\nu = \int_{-\infty}^{\infty} x \phi_{(-1,1)}(x) dx$. The last equality uses interchange of Lebesgue integral (w.r.t Lebesgue measure) with usual Riemann integral - since the Riemann integral over bounded intervals are finite and stays bounded as the interval grows. From basic probability calculations, using usual Riemann calculus, we know that this is same as the expectation of $Normal(-1, 1)$ random variable, which is -1.